
Lab 03: Build a Simple Website Endpoint with Azure Functions

AZ-900 Azure Portfolio

Objective

Create an Azure Function App, deploy a small HTTP-triggered function to it, test the public endpoint, require a function key to call it, and review the requests in Application Insights.

Lab environment

Resource group	rg-gp-functions-endpoint
Function App	func-gp-endpoint-65635750
Runtime	Node.js 22 LTS (JavaScript, Functions programming model V4)
Hosting plan	Flex Consumption (Linux)
Region	North Europe
Function	GetStatus (HTTP trigger)
Monitoring	Application Insights
Tools	Azure Portal, Azure Cloud Shell (Bash), Azure Functions Core Tools

Workflow

Create > Deploy > Test > Secure > Monitor > Clean up

What I did

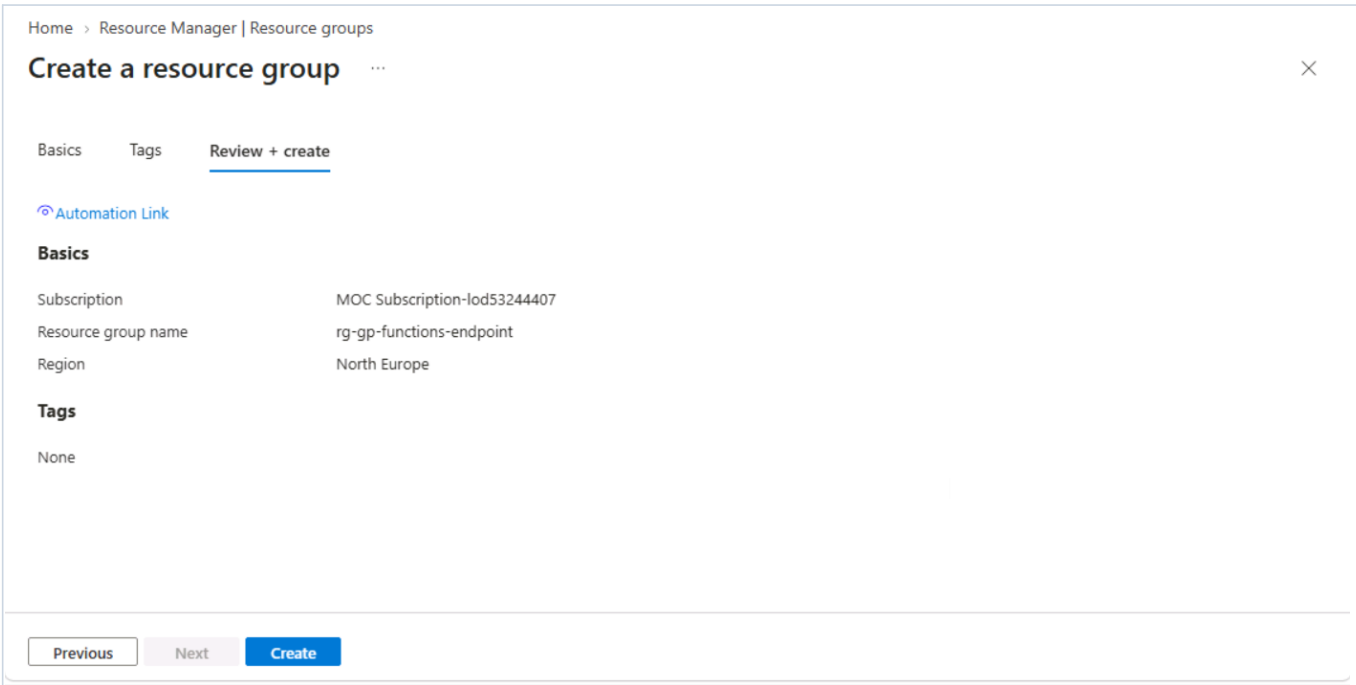
- **Create:** made the resource group and the Function App (Flex Consumption, Node.js 22 LTS) in North Europe.
- **Deploy:** in Cloud Shell, initialised a Node.js V4 project, added the `GetStatus` HTTP trigger with anonymous access, and published it with Azure Functions Core Tools. Azure returned an invoke URL.
- **Test:** called the public endpoint and checked the function in the Azure Portal.
- **Secure:** changed the authorization level from anonymous to function and redeployed. A request without a key returned HTTP 401. A request with the function key succeeded and returned *Hello, world!*
- **Monitor:** reviewed Application Insights, including Transaction Search and a Logs query.
- **Clean up:** deleted the resource group and monitoring resources, and removed the project folder from Cloud Shell.

Evidence note

The screenshots on the following pages cover resource group setup, deployment, the authorization change, the 401 response and the Application Insights query. Steps listed above that have no screenshot (the anonymous test, the successful keyed request, Transaction Search and cleanup) were completed during the lab but are not shown. The keyed request is left out on purpose because it would expose the function key. Screenshots that displayed the subscription ID were also left out.

1. Create: resource group

The resource group `rg-gp-functions-endpoint` was set up in North Europe with no tags. This screenshot is the Review + create step, before the Create button was pressed.



Home > Resource Manager | Resource groups

Create a resource group ...

Basics Tags **Review + create**

[Automation Link](#)

Basics

Subscription	MOC Subscription-lod53244407
Resource group name	rg-gp-functions-endpoint
Region	North Europe

Tags

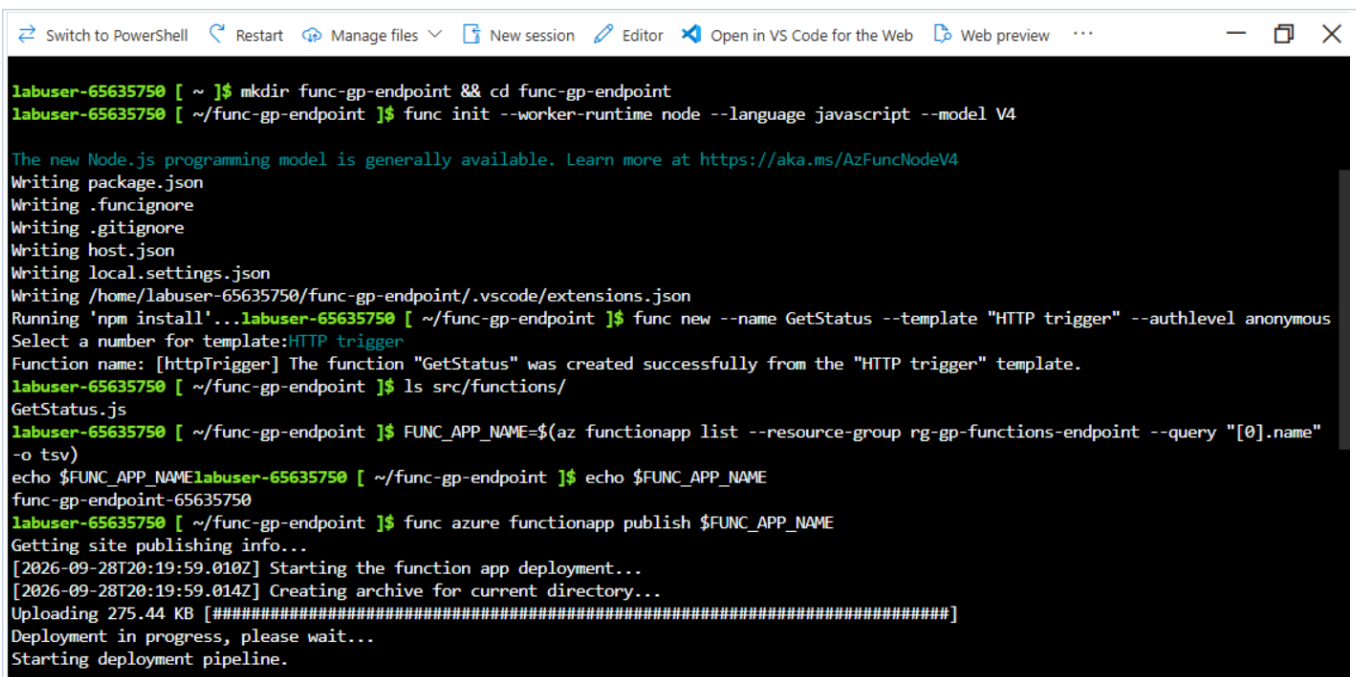
None

[Previous](#) [Next](#) [Create](#)

Figure 1. Resource group review page (Review + create).

2. Deploy: HTTP trigger from Cloud Shell

In Cloud Shell (Bash) I created the project folder, ran `func init` for Node.js, JavaScript and model V4, created the `GetStatus` HTTP trigger with anonymous auth, and started `func azure functionapp publish`. The screenshot ends while the deployment is still in progress.



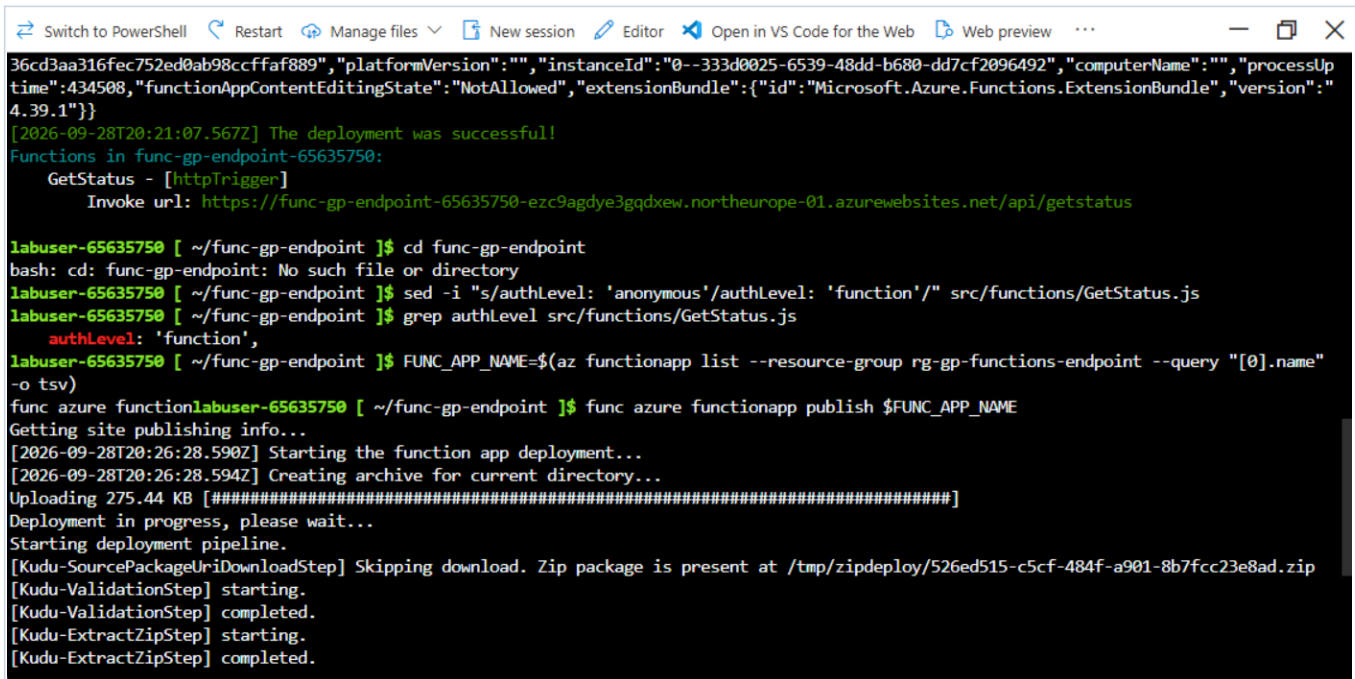
```
labuser-65635750 [ ~ ]$ mkdir func-gp-endpoint && cd func-gp-endpoint
labuser-65635750 [ ~/func-gp-endpoint ]$ func init --worker-runtime node --language javascript --model V4

The new Node.js programming model is generally available. Learn more at https://aka.ms/AzFuncNodeV4
Writing package.json
Writing .funcignore
Writing .gitignore
Writing host.json
Writing local.settings.json
Writing /home/labuser-65635750/func-gp-endpoint/.vscode/extensions.json
Running 'npm install'...labuser-65635750 [ ~/func-gp-endpoint ]$ func new --name GetStatus --template "HTTP trigger" --authlevel anonymous
Select a number for template:HTTP trigger
Function name: [httpTrigger] The function "GetStatus" was created successfully from the "HTTP trigger" template.
labuser-65635750 [ ~/func-gp-endpoint ]$ ls src/functions/
GetStatus.js
labuser-65635750 [ ~/func-gp-endpoint ]$ FUNC_APP_NAME=$(az functionapp list --resource-group rg-gp-functions-endpoint --query "[0].name" -o tsv)
echo $FUNC_APP_NAMElabuser-65635750 [ ~/func-gp-endpoint ]$ echo $FUNC_APP_NAME
func-gp-endpoint-65635750
labuser-65635750 [ ~/func-gp-endpoint ]$ func azure functionapp publish $FUNC_APP_NAME
Getting site publishing info...
[2026-09-28T20:19:59.010Z] Starting the function app deployment...
[2026-09-28T20:19:59.014Z] Creating archive for current directory...
Uploading 275.44 KB [#####]
Deployment in progress, please wait...
Starting deployment pipeline.
```

Figure 2. Project creation and start of deployment in Cloud Shell.

3. Secure: require a function key

The top of the screenshot shows the successful deployment and the generated invoke URL. I then used `sed` to change `authLevel` to 'function', confirmed it with `grep`, and started a redeploy. The screenshot ends partway through the redeploy. The `cd` error is a small typo on my part: I was already in the project folder.



```
Switch to PowerShell Restart Manage files New session Editor Open in VS Code for the Web Web preview ...
36cd3aa316fec752ed0ab98ccffaf889", "platformVersion": "", "instanceId": "0--333d0025-6539-48dd-b680-dd7cf2096492", "computerName": "", "processUp
time": 434508, "functionAppContentEditingState": "NotAllowed", "extensionBundle": {"id": "Microsoft.Azure.Functions.ExtensionBundle", "version": "
4.39.1"}}
[2026-09-28T20:21:07.567Z] The deployment was successful!
Functions in func-gp-endpoint-65635750:
  GetStatus - [httpTrigger]
  Invoke url: https://func-gp-endpoint-65635750-ezc9agdy3gqdxew.northeurope-01.azurewebsites.net/api/getstatus

labuser-65635750 [ ~/func-gp-endpoint ]$ cd func-gp-endpoint
bash: cd: func-gp-endpoint: No such file or directory
labuser-65635750 [ ~/func-gp-endpoint ]$ sed -i "s/authLevel: 'anonymous'/authLevel: 'function'/" src/functions/GetStatus.js
labuser-65635750 [ ~/func-gp-endpoint ]$ grep authLevel src/functions/GetStatus.js
  authLevel: 'function',
labuser-65635750 [ ~/func-gp-endpoint ]$ FUNC_APP_NAME=$(az functionapp list --resource-group rg-gp-functions-endpoint --query "[0].name"
-o tsv)
func azure functionlabuser-65635750 [ ~/func-gp-endpoint ]$ func azure functionapp publish $FUNC_APP_NAME
Getting site publishing info...
[2026-09-28T20:26:28.590Z] Starting the function app deployment...
[2026-09-28T20:26:28.594Z] Creating archive for current directory...
Uploading 275.44 KB [#####]
Deployment in progress, please wait...
Starting deployment pipeline.
[Kudu-SourcePackageUriDownloadStep] Skipping download. Zip package is present at /tmp/zipdeploy/526ed515-c5cf-484f-a901-8b7fcc23e8ad.zip
[Kudu-ValidationStep] starting.
[Kudu-ValidationStep] completed.
[Kudu-ExtractZipStep] starting.
[Kudu-ExtractZipStep] completed.
```

Figure 3. Deployment success, auth level change and redeploy.

4. Test: request without a key

After the change, opening the endpoint without a function key returns HTTP 401. This shows the endpoint is no longer open to anyone.

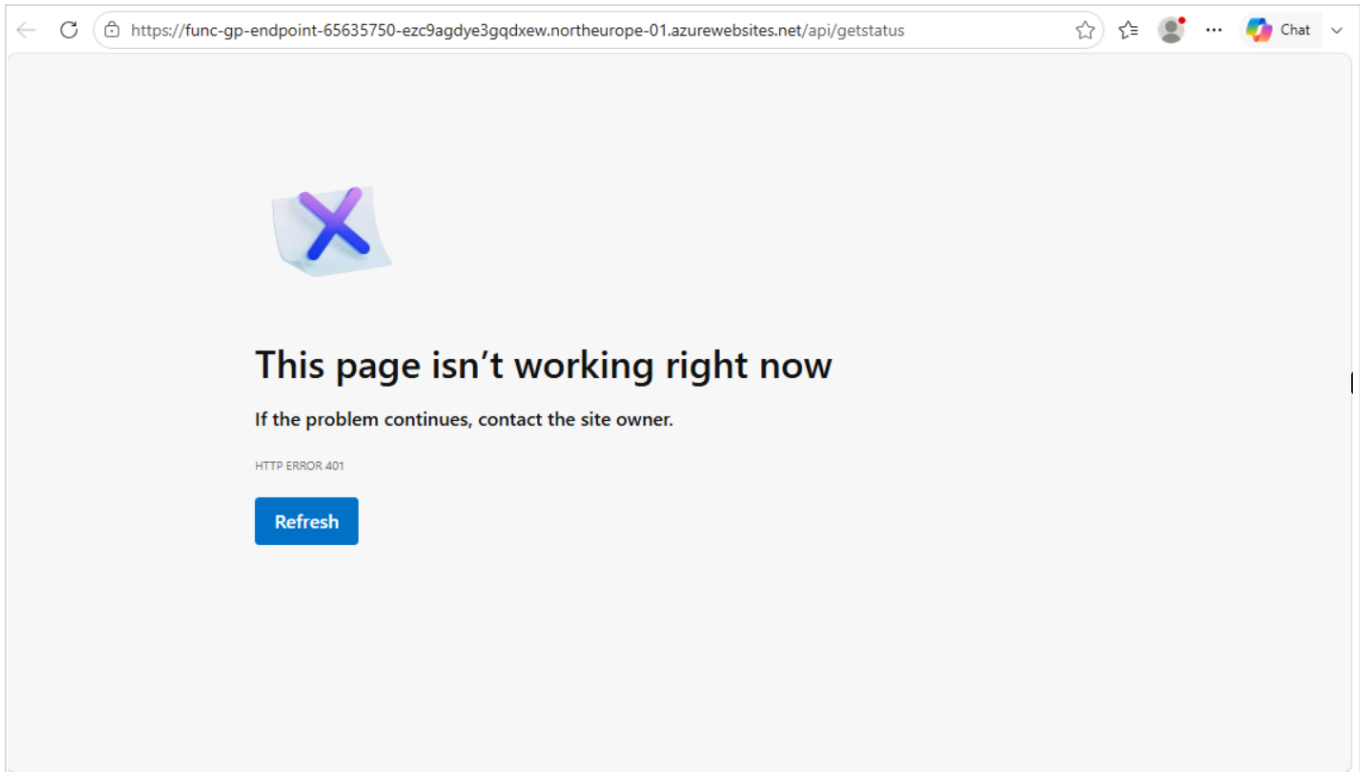
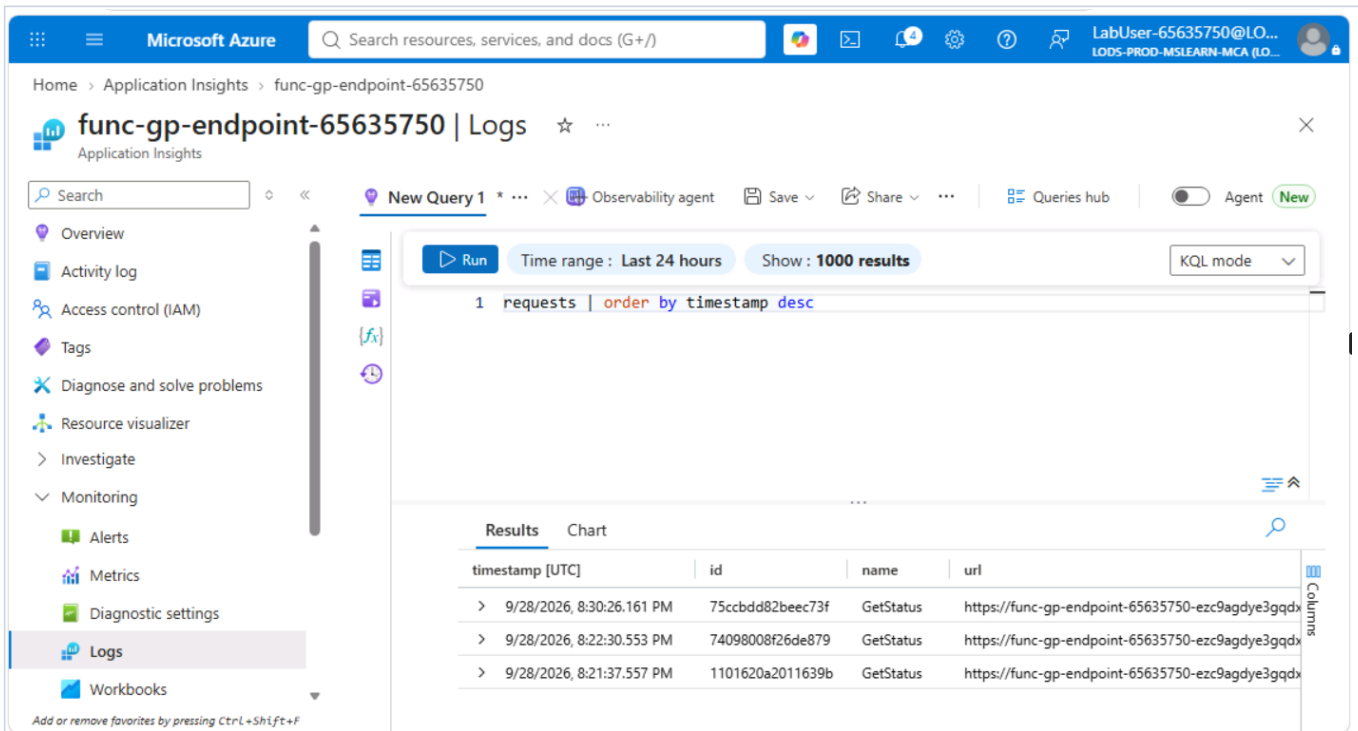


Figure 4. Browser request to the endpoint without a key returns HTTP 401.

5. Monitor: Application Insights Logs

In Application Insights I ran requests | order by timestamp desc over the last 24 hours. It lists three GetStatus requests. The result-code column is off-screen, so this screenshot does not show which were successful.



The screenshot shows the Microsoft Azure portal interface for Application Insights. The breadcrumb navigation is 'Home > Application Insights > func-gp-endpoint-65635750'. The page title is 'func-gp-endpoint-65635750 | Logs'. The left sidebar contains navigation links: Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Investigate, Monitoring, Alerts, Metrics, Diagnostic settings, Logs (selected), and Workbooks. The main area shows a 'New Query 1' editor with a 'Run' button, a 'Time range' of 'Last 24 hours', and 'Show: 1000 results'. The query is 'requests | order by timestamp desc'. Below the query editor, the 'Results' tab is active, showing a table with columns: timestamp [UTC], id, name, and url. The table contains three rows of data, all with the name 'GetStatus'.

timestamp [UTC]	id	name	url
> 9/28/2026, 8:30:26.161 PM	75ccbdd82beec73f	GetStatus	https://func-gp-endpoint-65635750-ezc9agdy3gqdx
> 9/28/2026, 8:22:30.553 PM	74098008f26de879	GetStatus	https://func-gp-endpoint-65635750-ezc9agdy3gqdx
> 9/28/2026, 8:21:37.557 PM	1101620a2011639b	GetStatus	https://func-gp-endpoint-65635750-ezc9agdy3gqdx

Figure 5. Application Insights Logs query showing GetStatus requests.

Key takeaways

- A Function App runs small pieces of code (functions) without me managing servers.
- An HTTP trigger gives a function a public URL, and the authorization level decides who can call it.
- anonymous allows anyone; function requires a key, and calls without one get 401.
- Application Insights records requests so I can check what happened after testing.
- Deleting the resource group removes everything inside it, which avoids leftover cost.